

ANCHORAGE BRYANT AAF, AK, USA

WMO: 702700

Lat: 61.266N

Lon: 149.653W

Elev: 388

StdP: 14.49

Time Zone: -9.00 (NAK)

Period: 86-19

WBAN: 00489

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.1	-12.1	-29.1	1.1	-16.7	-21.0	1.8	-11.2	25.3	39.4	20.2	36.2	1.5	30	0.641

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.6	75.7	60.8	72.6	59.4	69.3	57.7	62.2	73.5	60.2	70.6	58.6	67.5	4.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.8	69.7	66.0	55.3	66.0	62.9	54.3	63.5	62.0	28.0	72.8	26.6	71.4	25.5	67.5	68.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
19.5	15.2	11.8	-22.7	81.6	7.2	2.7	-27.9	83.5	-32.1	85.1	-36.1	86.6	-41.4	88.6		
			-23.0	64.9	7.1	1.9	-28.1	66.3	-32.2	67.4	-36.2	68.4	-41.3	69.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.3	15.1	15.8	25.3	36.8	47.9	56.0	59.5	56.5	46.9	34.8	20.7	18.3
	DBStd	18.53	13.84	12.67	10.26	6.74	4.83	4.40	3.74	4.26	6.89	8.51	12.19	11.83
	HDD50	5770	1083	957	765	395	99	1	0	3	131	473	880	983
	HDD65	10500	1548	1377	1230	845	530	274	177	263	542	936	1330	1448
	CDD50	754	0	0	0	0	34	180	294	206	38	2	0	0
	CDD65	10	0	0	0	0	0	3	6	1	0	0	0	0
	CDH74	184	0	0	0	0	1	52	111	20	0	0	0	0
	CDH80	24	0	0	0	0	0	5	19	0	0	0	0	0
	WSAvg	3.0	2.6	2.6	3.1	3.5	4.0	3.4	2.7	2.6	3.1	2.5	2.5	3.0
	PrecAvg	18.0	1.1	1.0	0.8	0.5	0.7	1.1	1.9	2.8	3.3	1.9	1.4	1.5
(15) Precipitation	PrecMax	23.3	2.2	2.4	2.0	1.8	1.7	2.6	3.6	6.7	6.9	4.6	3.0	2.7
	PrecMin	13.3	0.2	0.1	0.1	0.1	0.2	0.3	0.6	1.1	1.0	0.7	0.1	0.1
	PrecStd	2.4	0.6	0.6	0.4	0.4	0.4	0.6	0.8	1.4	1.6	0.9	0.7	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.9	44.1	46.9	56.8	72.5	79.9	82.2	76.9	65.1	56.5	46.5	46.8
		MCWB	37.6	37.6	39.2	45.3	55.3	61.2	63.9	62.2	54.4	47.4	39.0	39.0
	2%	DB	41.9	40.7	43.8	53.1	66.5	74.4	76.9	73.4	61.8	52.5	42.7	43.5
		MCWB	36.3	35.1	37.4	43.4	52.7	58.7	61.6	60.9	52.8	44.9	37.0	37.3
	5%	DB	38.6	37.1	41.4	50.6	62.6	70.3	73.2	69.7	59.3	49.7	38.6	39.0
		MCWB	34.0	33.2	36.0	41.6	50.9	57.4	60.1	58.5	51.7	43.6	35.8	35.0
	10%	DB	33.7	32.7	39.0	48.2	58.9	66.4	69.9	66.1	56.8	46.9	35.6	33.4
		MCWB	31.7	30.7	34.1	40.2	48.8	55.0	58.5	56.4	50.6	42.5	33.7	31.7
	0.4%	WB	38.9	38.7	40.2	46.9	56.3	63.0	65.6	64.8	56.3	48.9	40.0	39.3
		MCDB	43.6	43.3	45.3	55.0	70.1	75.2	82.3	74.6	61.4	53.8	44.2	44.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.4	35.6	38.2	44.3	53.6	60.1	62.4	61.8	54.2	46.2	38.0	37.1
		MCDB	41.3	40.0	42.9	52.1	65.0	73.0	74.8	71.2	59.1	50.4	41.5	42.4
	5%	WB	34.2	33.3	36.2	42.3	51.3	57.8	60.5	59.6	52.9	44.7	36.0	34.6
		MCDB	38.0	36.6	40.7	49.6	60.8	68.4	71.3	68.4	57.2	48.4	38.7	38.3
	10%	WB	31.7	30.6	34.6	40.6	49.4	55.5	59.1	57.7	51.4	43.1	33.6	31.8
		MCDB	34.4	33.1	38.6	47.3	57.9	65.8	68.5	64.2	55.6	46.7	35.4	33.5
(35) Mean Daily Temperature Range	5% DB	MDBR	14.3	18.2	19.0	18.5	18.2	17.4	16.6	15.7	14.3	13.7	13.5	13.2
		MCDBR	15.6	16.7	16.5	22.8	25.6	26.0	26.1	23.4	16.0	15.3	13.8	15.4
		MCWBR	11.7	12.6	11.7	15.0	15.0	14.5	13.8	12.2	9.4	9.6	9.8	10.7
	5% WB	MCDBR	15.0	15.2	16.0	20.3	23.5	23.3	22.8	20.0	13.9	13.2	12.9	14.5
		MCWBR	11.6	11.7	11.9	14.1	14.3	13.7	12.7	11.1	9.0	8.8	9.3	10.8
(40) Clear-Sky Solar Irradiance	taub		0.229	0.282	0.302	0.340	0.359	0.354	0.366	0.350	0.312	0.293	0.238	N/A
	taud		2.121	2.229	2.260	2.244	2.312	2.411	2.413	2.464	2.557	2.453	2.113	N/A
	Ebn at Noon		161	221	259	269	273	277	270	266	259	221	150	N/A
	Edh at Noon		12	21	28	35	36	33	32	28	22	17	11	N/A
(44) All-Sky Solar Radiation	RadAvg		91	311	822	1363	1643	1687	1506	1192	776	358	132	48
	RadStd		13	41	85	119	138	104	121	126	104	42	20	6

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (25 neighbors)	N/A	+3.56	+2.95	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page